DCV Project Report

Airline recovery analysis post COVID and other analysis

Project Members :
- Pranauv G : DA25B035
- Sashwath J S : DA25B043
- Subhranshu Das : DA25B050

LLM Tool Used :

- Claude (Anthropic) - Was used to verify and debug scraping and cleaning code. Helped with building of website, identifying errors and formatting issues at various points.

Contribution Split :

- Subranshu DA25B050 - Scraped data from DGCA website using selenium, Came up with certain visualisation
- Pranauv G DA25B035 - Introduced city-pair metrics and related visualisations. Also international flight data to improve data visualisation
- Sashwath J S DA25B043 - Scraped data from Github source, helped to come with metrics derived from data columns

Moreover, all 3 collaborated on data cleaning, improving graphs and other refinements.

For the project, data was collected from two sources the DGCA website and the Vonter GitHub

The DGCA data had to be cleaned and compared with the GitHub data to ensure correctness

Eight workbooks could not be fetched (HTTP 403): airasia17, go air25-26, trujet24-26 and vistara25-26.

No host serves a robots.txt, and DGCA's Website Policy says "Material featured on this website may be reproduced free of charge", provided the source is acknowledged.

Airline names come from file names, and placeholder cells ("-", spaces, "NR") become empty. Parsing stops on any other text in a number column.

All records matched with the Vonter copy agree on passengers, which checks the parser, not DGCA's own figures.

The storage file format is Parquet because it preserves dtypes, with a CSV copy of the parsed table for inspection

Cleaning Summary :

Step	Rows before	Rows after	Why
Remove empty template rows	5,844	3,034	DGCA lists every section for every airline, even when it did not fly
Keep domestic sections	3,034	2,191	International is analysed separately (536 usable rows)
Remove non-scheduled rows	2,191	1,426	Charters are out of scope
Set aside DGCA's Total Domestic	1,426	1,309	Used as an outside check, not as an airline
Stop at last reconciled month	1,309	1,301	Airlines add up to DGCA's total within 2% until August 2026
City pairs: merge into routes	66,448	65,764	194 city names become 144; both directions become one route

There are no exact duplicates or repeated airline-month keys. Brands are grouped - eg. Air India, Air India Express, Vistara, AirAsia India and AIX Connect become Air India Group

In the analysis, pre-COVID is treated as all periods for which we have data till March 2020 (ie) starting at April 2017

From columns of data, we are getting 7 metrics as shown in the dashboard linked at the end which help us to analyze the airlines in a better way

Based on the raw values of the metrics, we are standardizing them using a scoring method - Score = 50 + 25 x (value - median) / spread, capped at 0 to 100

An airline is scored only if it flew all 12 months with at least 25,000 passengers a month; seven qualify

To account for enormous growth by newer airlines (which is a characteristic of any new business popping up), growth is in log scale and momentum is only calculated until atleast 3 years of operation

Tests for Missingness

Kind of gap	Rows	Inference	Basis
Whole row empty	2,810	MNAR, structural: blank because the airline did not fly that section	How DGCA's sheets are built; not testable from the data
Load factor at zero traffic	94	Not truly missing: load factor is 0/0	Every load-factor-only gap is a zero-traffic row
Real gaps in reported domestic rows	11	Not MCAR. MAR on period is the best working assumption; MNAR cannot be ruled out	Exact test on period; see below
No report during a flight suspension	126 group-months	Structural: the value is known to be zero	Outside event

Fisher and T tests were performed to determine the missingness mechanism
We know from general information that a majority of the empty rows were because of COVID and when specific airlines were suspended

The structural gaps (due to COVID) are retained as they are crucial to the analysis

Limitations

- Scores are relative: 50 means typical of the airlines scored that month, not an absolute standard. Only seven airlines are scored.
- Operations metrics favour big jets on long trips; turboprop operators (Star Air, Fly91, Alliance Air) score low there regardless of management.
- The countermeasure settings (floors, 36-month maturity, 50 for a missing pillar) are judgment calls; the old-vs-adjusted chart shows their effect.
- City-pair data has no airline column, and some city merges are judgment calls (two Goa airports, Navi Mumbai into Mumbai, Hindon into Ghaziabad).
- Months with no scheduled international flights (April 2020 to early 2022) are treated as zero; air-bubble flights in that period are not in this series (to avoid them affecting general passenger flight data)
- DGCA data is provisional. Airline analysis stops at August 2026 (last month reconciling to DGCA's total within 2%); city-pair data ends July 2026.

Conclusions

Question	Result
Did traffic recover?	Passengers per month are 25% above the pre-COVID average and passenger-km 28% above. Of the five airlines, four regained their level (IndiGo December 2021; Air India Group and Alliance Air April 2022; Go First May 2022), two are still above it, and SpiceJet never got back.
Are planes as full?	Mostly not: only Air India Group fills more seats than before COVID (+1.6 points; IndiGo -2.5, SpiceJet -7.7). Capacity came back faster than demand.
Who gained?	IndiGo's share of passenger-km rose from 45% to 62%, and the concentration index (HHI) from about 2,840 to about 4,680, where above 2,500 is highly concentrated.
Where did growth come from?	Smaller cities: routes between them are at 145% of 2019, against 100% for metro-to-metro. The busiest 10% of routes carry 65% of passengers (66% in 2019), and 367 routes are new. 8 of 46 large cities are still below 2019, including Kannur (-43%) and Kolkata (-8%).
And international?	Well above pre-COVID, driven by IndiGo at 363% of its level (178% at home). Air India Group is at 139% and SpiceJet at 46%.
Who performs best?	IndiGo 61, Akasa Air 57, Air India Group 54, down to Alliance Air 12. Size is not the whole story: Akasa Air ranks second as a new entrant.
What predicts growth?	Past growth does (+0.75), then the Recovery pillar (+0.58); the Operations pillar does not (+0.01). Go First ranked only 3 of 5 (index 40) in its last full month, so traffic data alone did not flag its collapse. A simple seasonal projection of market traffic misses by 4.6% to 6.4%, one to six months ahead.

Dashboard Link

Deployed through GitHub pages

Link : https://iitm-da25b035.github.io/DA2402-Project-Deploy